

Week 1: Requirement Analysis and Customer Database Module

1. Requirement Analysis

Understanding Pet Shop Business

A pet shop management system handles the day-to-day activities of a pet shop, including customer management, pet sales, product sales, inventory, suppliers, employees and billing. Below is a structured description of the business process, customer data requirements and entity details for database design.

The Process Flow:

- **Customer Registration / Visit:** The customer provides basic details before purchasing a pet or product.
- **Browse Pets / Products:** The customer checks the available pets, food, toys, medicines and accessories.
- **Select Item:** The customer chooses the required pet or product.
- **Billing / Payment:** The purchase is recorded and the customer completes the payment.
- **Stock Update:** After the sale, the quantity of the product or availability of the pet is updated.
- **Support / Feedback:** The customer can contact the shop for product support, returns or other queries.



Entities:

Entity	Description	Key Attributes	Key Relationships
Customer	Person who purchases pets or products from the shop.	customer_id (PK), first_name, last_name, email, phone, created_at	One-to-Many with Sale
Pet	Animal available for sale in the pet shop.	pet_id (PK), pet_name, species, breed, age, price, health_status	Many-to-One with Customer/Sale
Product	Pet food, toys, medicines and accessories sold by the shop.	product_id (PK), product_name, category, price, stock_quantity, supplier_id (FK)	Many-to-One with Supplier
Sale	Transaction representing a customer purchase.	sale_id (PK), customer_id (FK), sale_date, total_amount, payment_method	Many-to-One with Customer
Supplier	Person or company supplying products to the shop.	supplier_id (PK), supplier_name, email, phone, address	One-to-Many with Product

Employee	Staff member responsible for shop activities.	employee_id (PK), employee_name, phone, role, salary	Manages shop operations
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Customer Table Creation

The Customer table stores the basic information of customers who purchase pets or products from the pet shop.

```
SQL> create table customer(customer_id varchar2(30) primary key, first_name varchar2(30), last_name varchar2(30), email varchar2(30), phone number(10), created_at date);
Table created.
```

```
SQL> desc customer;
Name                                Null?    Type
-----
CUSTOMER_ID                        NOT NULL VARCHAR2(30)
FIRST_NAME                          VARCHAR2(30)
LAST_NAME                          VARCHAR2(30)
EMAIL                              VARCHAR2(30)
PHONE                              NUMBER(10)
CREATED_AT                          DATE
```

Insert Values:

```
SQL> insert into customer values('S101','Akshaya','Nivasini','akshayanivasinikr@gmail.com',9791706780,TO_DATE('2024-06-13','YYYY-MM-DD'));
1 row created.
```

```
SQL> insert into customer values('S102','Prithish','Raj','prithishraj@gmail.com',9791345610,TO_DATE('2025-07-11','YYYY-MM-DD'));
1 row created.
```

```
SQL> insert into customer values('S103','kshaunish','Rajesh','kshaunishrajesh@gmail.com',9791345234,TO_DATE('2024-01-01','YYYY-MM-DD'));
1 row created.
```

```
SQL> insert into customer values('S104','kaniska','Ram','kaniskaram@gmail.com',9123455234,TO_DATE('2025-12-14','YYYY-MM-DD'));
1 row created.
```

```
SQL> insert into customer values('S105','Rukmani','Kuppusamy','Rukmanisamy@gmail.com',7143455234,TO_DATE('2020-09-24','YYYY-MM-DD'));
1 row created.
```

```
SQL> insert into customer values('S106','Vijay','Deepika','Vijideepi@gmail.com',7112355234,TO_DATE('2022-05-02','YYYY-MM-DD'));
1 row created.
```

```
SQL> insert into customer values('S107','Aura','Nivas','Auranivas@gmail.com',7123456789,TO_DATE('2018-02-07','YYYY-MM-DD'));
1 row created.
```

```
SQL> insert into customer values('S108','Shreya','Raman','Shreyaraman@gmail.com',7987654219,TO_DATE('2019-08-21','YYYY-MM-DD'));
1 row created.
```

```
SQL> insert into customer values('S109','Geetha','Govind','Geethagovind@gmail.com',7988763451,TO_DATE('2019-05-05','YYYY-MM-DD'));
1 row created.
```

```
SQL> insert into customer values('S110','Ramya','Krishnan','ramyakrish@gmail.com',7118763451,TO_DATE('2017-08-05','YYYY-MM-DD'));
1 row created.
```

Like this, we can insert as many customer records as required.

```
SQL> select * from customer;
```

CUSTOMER_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE	CREATED_A
S101	Akshaya	Nivasini	akshayanivasinikr@gmail.com	9791706780	13-JUN-24
S102	Prithish	Raj	prithishraj@gmail.com	9791345610	11-JUL-25
S103	kshaunish	Rajesh	Kshaunishrajesh@gmail.com	9791345234	01-JAN-24
S104	kaniska	Ram	Kaniskaram@gmail.com	9123455234	14-DEC-25
S105	Rukmani	Kuppusamy	Rukmanisamy@gmail.com	7143455234	24-SEP-20
S106	Vijay	Deepika	Vijideepi@gmail.com	7112355234	02-MAY-22
S107	Aura	Nivas	Auranivas@gmail.com	7123456789	07-FEB-18
S108	Shreya	Raman	Shreyaraman@gmail.com	7987654219	21-AUG-19
S109	Geetha	Govind	Geethagovind@gmail.com	7988763451	05-MAY-19
S110	Ramya	Krishnan	ramyakrish@gmail.com	7118763451	05-AUG-17

```
10 rows selected.
```

Update:

```
SQL> update customer set phone = 9791706669 where customer_id = 'S101';
```

```
1 row updated.
```

Customer Database Module

The customer database module maintains customer details and supports basic operations such as inserting new records and updating existing information.